

Beyond the Prompt: Cultivating Intrinsic Motivation in the Generative Renaissance

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Abstract

Generative artificial intelligence (AI) is reshaping computer graphics and interactive techniques, sparking a “Generative Renaissance” that transforms how students create, learn, and collaborate. While these tools expand access to powerful creative capabilities, they also raise critical questions for educators: How can learning remain meaningful when a prompt can generate polished code, images, or text? How can curricula shift from performance-based incentives to cultivating deep, resilient, intrinsic motivation?

This panel brings together speakers from academia, industry, and policy to share innovative research outcomes, methods, and developments in the education of computer graphics and interactive techniques. Panelists will explore human–AI synergy in learning, highlighting projects that integrate generative AI into multimedia quality assessment, three-dimensional (3D) reconstruction, visual computing, language technologies, and interdisciplinary curricula. They will present strategies for project-based education that combine computational methods, aesthetic exploration, and experiential learning—including industry collaborations, fieldwork at innovation hubs, and community-engaged initiatives.

Together, the panel will address key questions:

- How can assignments reward inquiry and creativity over AI-generated polish?
- What “power skills” such as ethical reasoning, critical inquiry, and collaboration emerge from human–AI co-creation?
- What frameworks ensure equitable, ethical, and globally relevant adoption of generative tools in education?

With moderators bridging academia and industry, this panel directly supports the Educator’s Forum mission to advance innovative pedagogy in computer graphics and interactive techniques. Attendees will gain globally informed strategies to inspire intrinsically motivated learners—students who are not merely AI users, but discerning, creative collaborators in the Generative Renaissance.

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CCS Concepts

• **Human-centered computing** → **Collaborative and social computing design and evaluation methods**; • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Interactive learning environments**; • **Social and professional topics**;

Keywords

Computer Graphics; Interactive Techniques; Generative AI; Intrinsic Motivation; Diversity, Equity, and Inclusion (DEI); Sustainable Development Goals (SDGs); Curriculum Design; Human–AI Collaboration; Virtual Reality (VR); Extended Reality (XR); Natural Language Processing (NLP); High-Performance Computing (HPC); Artificial Intelligence-Generated Content (AIGC); Interdisciplinary Education

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1 Overview

Computer graphics and interactive techniques—including rendering, modeling, animation, visualization, and interactive media such as virtual reality (VR) and extended reality (XR)—have long served as central domains through which students learn to think critically, create, and collaborate [Franco et al. 2006]. With the emergence of generative artificial intelligence (AI), these traditions are undergoing rapid transformation. Generative AI enables new forms of prototyping, multimodal synthesis, and automated design iterations [Sharma et al. 2025]. This integration introduces both significant challenges and promising opportunities for education in computer graphics and interactive techniques [Basole and Major 2024].

The primary challenge lies in preventing students from relying on AI to perform creative labor in ways that diminish their intrinsic motivation to learn [Westgeest 2025]. Conversely, the opportunity is to leverage AI as a pedagogical catalyst for deeper engagement. By carefully embedding AI tools into curricula, educators can encourage students to reflect critically on inquiry, creativity, and ethical responsibility as central elements of the learning process [Baruah and Paulus 2018; Csikszentmihalyi 1990; Kayyali 2024]. In

this way, generative AI can function not as a substitute for student effort, but as a scaffold that foregrounds reflective practice, iterative design, and critical thinking.

Importantly, educators can redirect attention toward process-based evidence of learning, such as inquiry framing, cycles of iteration, and ethical reasoning. One promising avenue is the integration of the United Nations’ Sustainable Development Goals (SDGs), a global framework established in 2015 to address poverty, inequality, climate change, and education [United Nations 2015]. Embedding SDG themes into graphics and interactivity curricula enables students to situate their visualization work within broader social challenges, encouraging them to view technical accomplishments as contributions to global responsibility.

2 Literature Review

The cultivation of intrinsic motivation has long been central to educational research. Flow theory [Csikszentmihalyi 1990] demonstrates that learners remain engaged when challenge, autonomy, and feedback are in balance. Gagné’s learning hierarchies [Gagne 1968] emphasize the satisfaction that arises from progression through increasingly complex skills. Creative education traditions highlight ideation techniques that make thinking visible, thereby promoting student ownership and reflective habits of mind [Young and Reinhard 1975]. Collectively, these perspectives underscore the importance of designing educational environments that cultivate reflection, autonomy, and deep engagement.

Past contributions to the SIGGRAPH Asia Educator’s Forum demonstrate how these theoretical principles have been operationalized in computer graphics and interactive techniques education. For instance, stereoscopic education materials emphasized reflection on learning processes rather than polished outputs [Miyai and Yamaguchi 2015], showing how evaluation can privilege process over product. VR group projects cultivated collaboration and design literacies [See et al. 2023], providing evidence that teamwork and co-creation can enhance engagement and mitigate disengagement risks. Virtual production pipelines connected artistic expression with computational methods [Bennett et al. 2023], offering a model for embedding emerging technologies into project-based workflows that require human judgment and iteration. XR curricula created inclusive pathways for learners with limited programming backgrounds [Marks and Gil Parga 2023], illustrating how technological innovation can broaden participation while maintaining critical inquiry and reflection.

Each of these contributions embodies strategies—reflection, collaboration, inclusivity, and responsibility—that directly cultivate intrinsic motivation. These strategies provide clear precedent and scaffolding for addressing the dual challenge and opportunity introduced by generative AI: preventing disengagement while fostering deeper, more socially engaged learning. A sample rubric (Table 1) illustrates how these strategies can be operationalized in graphics education, while also leaving space for open questions that will be addressed in the panel discussion.

At the same time, generative AI introduces new tensions into this pedagogical landscape. Because AI systems can generate high-quality artifacts instantly, extrinsic markers such as visual polish

Table 1: Rubric linking intrinsic motivation strategies to graphics education. Dimensions represent core pedagogical values; columns show evidence levels in student work.

Dimension	Low	Medium	High
Reflection	✗ Minimal or no documentation of decisions; limited awareness of process.	● Some rationale given; reflective notes appear but lack depth or consistency.	✓ Systematic process diaries, critique responses, and transparent decision-making logs.
Collaboration	✗ Individual-only work; no coordination or shared assets.	● Intermittent peer input; limited collaborative artifacts.	✓ Active teamwork with shared design decisions and traceable contributions.
Inclusivity	✗ Narrow scope; no attention to accessibility or diversity.	● Acknowledges diverse users but minimal adaptation.	✓ Deliberate inclusive practices; accessible design addressing varied user needs.
Responsibility (SDGs) 	✗ No societal or ethical consideration.	● General awareness of broader social context.	✓ Explicit integration of United Nations SDGs with justification of societal impact.

Legend: ✗ = Not evident, ● = Emerging, ✓ = Strong evidence. Header colors align with the workflow stages: Low/Exploration, Medium/Analysis, High/Responsibility.

Open questions for moderated discussion: (1) How can assignments reward inquiry and creativity over AI-generated polish? (2) What “power skills” (ethical reasoning, critical inquiry, collaboration) should be evidenced in graphics projects? (3) What frameworks ensure equitable, ethical, and globally relevant adoption of generative tools in education?

lose diagnostic value [Zhao et al. 2024]. Yet these same tools also afford opportunities for scaffolding, critique, and iterative refinement [Feng et al. 2024]. Figure 1 therefore emphasizes a process-based framework that aligns with the rubric in Table 1, ensuring that graphics and interactivity education remains a site for cultivating resilient, intrinsically motivated, and socially responsible learners while explicitly connecting student work to responsibility and higher purpose.

3 Panel Description

This panel convenes experts from academia, industry, and policy whose diverse backgrounds collectively illustrate the breadth of computer graphics and interactive techniques education in the era of generative artificial intelligence (AI). Dr. Dongping Liu (Amazon Web Services) contributes an industry perspective by presenting AWS’s open-source data and software platforms that support education in computer graphics and interactive techniques, emphasizing equitable access to resources that democratize participation in digital transformation. Prof. Xiaojuan Qi (University of Hong Kong) brings expertise in computer vision and three-dimensional (3D) reconstruction, demonstrating how generative models and interpretability can be embedded into graphics curricula to foster inquiry and technical depth. Prof. Shiqi Wang (City University of Hong Kong) offers leadership in multimedia AI and perceptual quality assessment, showing how evaluation of AI-generated content can support reflective learning and process-oriented assessment. Prof. Kam Fai Wong (Chinese University of Hong Kong) integrates technical innovation in natural language processing (NLP) with policy experience, highlighting the importance of multilingual and cross-cultural communication for inclusive participation in interactive technologies. Prof. Luyao Zhang (Duke Kunshan University) contributes a distinctive interdisciplinary approach that integrates

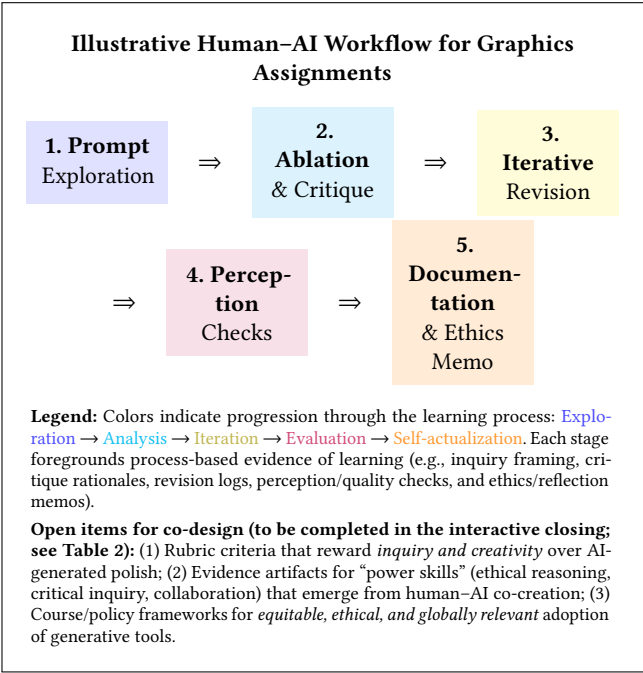


Figure 1: Process-first blueprint that foregrounds inquiry, critique, iteration, perception checks, and documentation. The figure operationalizes the literature-driven shift from product polish to process evidence and sets up the panel’s co-design of a detailed rubric and workflow.

computer graphics, generative AI, and social sciences into project-based pedagogy, explicitly linking student learning to the United Nations Sustainable Development Goals (SDGs) and cultivating responsibility and global awareness.

Together, these speakers embody the Educator’s Forum mission by advancing diversity, equity, and inclusion (DEI) while directly addressing the pedagogical challenges and opportunities identified in the Overview and Literature Review. The panel highlights inclusive curricula that broaden participation in graphics and extended reality (XR) education [Marks and Gil Parga 2023], cross-cultural frameworks that strengthen digital citizenship, equitable infrastructures that democratize access through open-source platforms, and interdisciplinary curricula that connect graphics education to pressing global challenges via the SDGs. Through these complementary perspectives, the session creates a safe and collaborative space for dialogue, knowledge sharing, and critical reflection aimed at advancing computer graphics and interactive techniques education worldwide.

3.1 Speaker Biographies

Dr. Dongping Liu. Dr. Liu is Senior Industry Business Development Manager, Higher Education & Research at Amazon Web Services. He received a Ph.D. in Physics from the Chinese Academy of Sciences and completed postdoctoral research at McGill University. He previously served as Associate Researcher at the Chinese Academy of Sciences. His expertise spans artificial intelligence (AI), deep

learning, and high-performance computing (HPC) for educational digital transformation. He is a recipient of the Beijing Science & Technology Award and has authored over thirty publications and holds more than ten patents. He is also Chairman of the Hong Kong branch of the China Computer Federation (CCF) Young Computer Scientist and Engineer Forum (YOCSEF).¹

Prof. Xiaojuan Qi. Prof. Qi is Assistant Professor in the Department of Electrical and Electronic Engineering at the University of Hong Kong. She is an accomplished researcher in computer vision, 3D reconstruction, and AI applications. Her work has been published at major venues including CVPR, ICCV, and ECCV. Her projects emphasize real-world visual AI systems for autonomous driving and embodied agents.²

Prof. Shiqi Wang. Prof. Wang is Associate Dean of the College of Computing and Professor in the Department of Computer Science at City University of Hong Kong. He is a leading expert in multimedia AI and quality assessment, with over 400 publications and an h-index of 64. He is the recipient of the IEEE Multimedia Rising Star Award. His research focuses on artificial intelligence-generated content (AIGC), perceptual quality assessment, and multimedia optimization.³

Prof. Kam Fai Wong. Prof. Wong is Professor in the Department of Systems Engineering and Engineering Management at the Chinese University of Hong Kong and serves as Associate Dean (External Affairs) in the Faculty of Engineering. He is also an elected member of the Hong Kong Legislative Council. An Association for Computational Linguistics (ACL) Fellow, he has published over 260 works on natural language processing (NLP), database systems, and cross-cultural communication technologies. His experience bridges technical innovation and policy-making.⁴

Prof. Luyao Zhang. Prof. Zhang is Senior Research Scientist at the Digital Technology Innovation Center and Assistant Professor of Economics at Duke Kunshan University. She co-leads the Center for Contemporary China Digital Technology and Society Cluster. Her research appears in venues such as CSCW, NeurIPS, ACM CCS, ACM S&P, and Nature Scientific Data. She pioneers project-based pedagogy that integrates generative AI, graphics, and interdisciplinary inquiry into curricula connected to the SDGs. Her teaching emphasizes experiential learning through collaborations with industry, innovation hubs, and community-engaged projects.⁵

3.2 Panel Structure with Integrated Illustrative Artifacts

The session is organized to progressively build from framing the challenges and opportunities of generative AI in computer graphics and interactive techniques, to lightning talks from diverse perspectives, to a structured discussion around key questions, and finally to an interactive co-design exercise with the audience. Table 2 presents the detailed agenda.

¹<https://www.linkedin.com/in/dpliu/>
²<https://xjq.github.io/>
³<https://www.cs.cityu.edu.hk/~shiqwang/>
⁴<https://www.se.cuhk.edu.hk/people/academic-staff/prof-wong-kam-fai/>
⁵<https://scholars.duke.edu/person/luyao.zhang>

Table 2: Panel agenda. Colors and icons highlight the flow of the session.

00–05 min	 Opening: Framing generative AI as both a challenge and opportunity for education in computer graphics and interactive techniques.
05–30 min	➤ Lightning Perspectives (5 minutes each): – Prof. Shiqi Wang <i>Perception-Driven Rubrics for AI-Generated Content</i> How quality assessment fosters reflective learning and process evaluation. – Prof. Xiaojuan Qi <i>Scaffolds from 3D Reconstruction to Generative Workflows</i> How staged inquiry and interpretability support deeper engagement. – Prof. Kam Fai Wong <i>Multilingual Inclusion and Digital Citizenship</i> How NLP and cross-cultural communication broaden access and collaboration. – Dr. Dongping Liu <i>AWS Open-Source Platforms for Graphics Education</i> How cloud-based data and software democratize access and ensure equitable participation. – Prof. Luyao Zhang <i>Interdisciplinary Education and SDG-Linked Curricula</i> How project-based pedagogy integrates AI, graphics, and self-actualization for global challenges.
30–50 min	❖ Moderated Discussion: Panelists and audience reflect on three guiding questions: (1) How can assignments reward inquiry and creativity over AI-generated polish? (2) What “power skills”—such as ethical reasoning, critical inquiry, and collaboration—emerge from human–AI co-creation? (3) What frameworks ensure equitable, ethical, and globally relevant adoption of generative tools in education?
50–60 min	 Interactive Blueprinting: Panelists and audience co-develop a draft rubric (Table 1) and workflow (Figure 1) as a blueprint for future research and pedagogical development.

4 Conclusion

Generative artificial intelligence (AI) challenges computer graphics and interactive techniques education to move beyond polished outputs toward cultivating inquiry, reflection, and responsibility. The significance of this panel lies in uniting diverse perspectives from academia, industry, and policy to reframe AI not as a substitute for learning, but as a catalyst for deeper motivation and socially engaged practice.

By drawing on established traditions of reflection, collaboration, inclusivity, and responsibility, the session highlights how innovation in pedagogy can be advanced through interdisciplinary dialogue. The concrete takeaway will be a co-designed set of rubrics and workflow frameworks that provide educators with actionable tools for aligning student learning with critical inquiry, ethical reasoning, and the United Nations Sustainable Development Goals

(SDGs). In doing so, the panel supports the Educator's Forum mission to foster innovation, equity, and global responsibility in computer graphics and interactive techniques education.

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